

BrainBlast Product Review

Status 26 Sep 2026: every finding below is fixed in v1.1 and locked in by regression tests (1,079 tests passing).

UX and flow, functional logic, answer checking, and the correctness of every maths and literacy item. The code, numbers and item ids were verified against the repo. The visual/screenshot pass was skipped.

5
Critical

23
High

40
Medium

52,500
maths questions generated

32,398
maths answers verified independently

983
literacy items read by hand

~43k
MC questions probed; all have exactly 1 correct option

Fix first

1. Custom spelling is unanswerable. "[listen carefully]" is shown, but nothing speaks the word. Auto-speak it, add replay, and show a clue sentence.
2. Stuck children have no way forward. Reveal the answer and explanation after 2 wrong tries, then continue. The same fix stops tap-every-option guessing from being praised.
3. Make isCorrect more tolerant. Accept curly apostrophes, Unicode minus, 9:20 vs 09:20, 7 / 8, 3 : 2, 78p vs 0.78, and check units. Right now correct children are being marked wrong.
4. Fix the wrong or broken maths items. HARD percentages give a rounded answer without saying so (41% of cases), the Fibonacci prompt contains a draft, 1/2 and 5/10 are both offered as options, and the float 1000.0000000000001 mℓ is shown.
5. Fix the grammar items with two defensible answers. gs540, gs530, ctx28, gs582, pv106/126, gs570.
6. Fix the Times Table timeout. It still costs two questions: timeLeft is never reset.
7. Stop Run & Learn serving passage questions without the passage. This affects 6–13% of grammar and vocab gates.
8. Confirm before leaving a round, and save the session so a refresh or accidental back doesn't lose it.
9. Validate Restore. review:{"x":null} crashes the Hub on every load, and a sibling's profile is overwritten without asking.
10. Match pronouns to names. 1,862 prompts pair a name with the wrong pronoun ("Callum ... change does she get?").

1 · UX, flow & consistency child + parent perspective

SEV	WHERE	PROBLEM	FIX
CRIT	spelling.js:218, session.js:158	The custom-word prompt never speaks the word, and TTS only reads the placeholder. These questions can be up to half of a spelling round.	Auto-speak the word, add a replay button, and add a clue sentence as a fallback.
CRIT	Quiz.jsx:171-181, 373	No reveal and no skip. After the first free hint, hints cost coins. A child with no coins who is stuck on a typed question can only leave with Back, which loses the round.	Offer "Show me" after 2 misses: show the answer and explanation, then Next.
HIGH	Quiz.jsx:16-24, 184, Game.jsx:114	Tapping every MC option earns "You worked that out" plus the coin sound.	Reveal after 2 wrong picks; use different praise and no coin sound on a retry.
HIGH	App.jsx:294, 357	Back leaves a round with no confirm, and a refresh loses it. An abandoned exam saves no history, but its answers still count towards mastery.	Add a "Leave this round?" dialog and keep the session in storage.
HIGH	TimesTable.jsx:108,181	Times Tables pays 1 coin per answer on any table (including ×2 and ×10), about 20 coins in 2 minutes. That beats a practice round.	Cap Times Tables coins per day, or pay only on tables not yet secure.
HIGH	App.jsx:332, Game.jsx:288	Run & Learn pays +5 for finishing whatever the score, and the win screen doesn't show it.	Scale the bonus by first-try accuracy and show it.
HIGH	screens.jsx:525-540, garden.js:17	The whole shop costs 412 coins and every item is one-off. Items are only ever shown inside the shop. The garden has no screen of its own and stops at stage 9.	Show collected items in a room or garden scene; add repeatable or seasonal rewards.
HIGH	Exam.jsx:197, Game.jsx:230, speech.js:62	No read-aloud in Exam, the game gates, passages, hints or "Why". "____" is not cleaned up before speaking.	One shared speak button on every text block; read blanks as "blank".

SEV	WHERE	PROBLEM	FIX
HIGH	screens.jsx:436, Backup.jsx:95	No parent gate. A child can pin Easy, turn the timer off, erase the profile, restore a save, or hand-edit the plain-text save to change coins.	Put a grown-up question in front of Settings, Erase and Restore.
HIGH	screens.jsx:278–285	The progress heading says "Keeps catching <i>her</i> out". Chips show raw ids like spot:sp12 and vocab:... . There are no dates and no trend.	Use the profile name, readable labels, and a 7-day activity list.
MED	screens.jsx:20, Hub.jsx:62,100	The welcome screen leaves out Vocabulary and says hints "cost a few" coins even though the first is free. The Hub offers about 12 actions with no first step, and has two identical "Start" buttons.	Give new profiles one "Start your first round" card.
MED	App.jsx:223–279	Welcome has no back button, and a back gesture from it lands on a hub with an empty name ("Hi 🖐").	Add a back button; handle Welcome in popstate.
MED	App.jsx:303	"Another round" after a topic round starts a mixed round, and after the daily challenge it just goes Home.	Repeat the same topic; relabel the button after the daily.
MED	App.jsx:111, Hub.jsx:53	Only the daily challenge advances the streak, and the daily card vanishes once done.	Count any finished round; show "Done today ✓".
MED	Quiz.jsx:33 vs Game.jsx:7	Hint pricing differs: 3 coins with a free first hint in Quiz, 5 coins with none free in the game.	Use one rule everywhere.
MED	exam.js:9, Exam.jsx:210	20–30 questions over 15–22 minutes is long for this age. Enter submits instantly with no review step.	Add a 10-question paper and a "Review answers" step.
MED	screens.jsx:434	Switch profile is hidden inside Settings.	Add an avatar button to the Hub header.
MED	TimesTable.jsx:281	Keypad only, no physical keyboard. The 5-second timer ignores the Settings timer toggle.	Add keydown handling and a relaxed mode.

SEV	WHERE	PROBLEM	FIX
MED	Quiz.jsx:197	When time runs out the child stays on the question, marked wrong, with an empty bar still showing.	Hide the bar after timeout.
LOW	various	Inconsistent wording: Practise/Practice, round/quiz/paper/test, 🔄/↺, Home/Back home, Keep/Delete vs Keep it/Erase, "Tic-Tac-Toe" (should be Noughts and crosses), "she runs automatically".	Keep a single copy glossary.
LOW	index.html, screens.jsx:397	The dyslexia font (Atkinson) is never loaded. Speed options are emoji only, so they are read aloud as "turtle". The canvas game ignores dyslexia mode and reduced motion.	Load the font and add text labels.
LOW	Hub.jsx:28, screens.jsx:252, 511	Empty states: new users see "Nothing to practise again today", Progress shows 0% accuracy, the shop gives no "need X more", and nothing explains 4 full profiles. Delete is a faint 🗑️ with no backup offer.	Write copy for each empty state.

2 • Functionality & logic

SEV	WHERE	PROBLEM	FIX
HIGH	TimesTable.jsx:93	The previous fix is incomplete. After a timeout, timeLeft stays at 0, so the next question fails the moment it appears. One timeout costs two questions and resets the streak.	Call setTimeLeft(TIMER_SEC) in the 420ms callback and in startGame .
HIGH	Game.jsx	Run & Learn asks "According to the passage..." without showing the passage. This is about 6% of grammar and 13% of vocab questions.	Filter out questions with q.passage , or render PassagePanel.

SEV	WHERE	PROBLEM	FIX
MED	session.js:73, 315	Orphan review keys (items no longer in the bank) are never graded, so they stay due forever. They fill all 4 review slots, leaving 0 real reviews.	Drop unresolvable keys before <code>slice()</code> , or prune them when loading.
MED	session.js:95	A passage cluster is added after the same items were already added as reviews, so they appear twice. This happened in 36 of 99 rounds with due passage items.	Skip the cluster if any of its keys are already in <code>usedKeys</code> .
MED	review.js:40	Every correct answer promotes the item, even when it isn't due. Five percentage questions in one round make the topic "mastered", and it never comes back.	Promote only when <code>due ≤ today</code> .
MED	session.js:69	The pinned-topic review filter doesn't work. Per-item keys (<code>grammar:cl151</code>) never match <code>grammar:clauses</code> , so pinned literacy topics get no reviews, and <code>me1</code> matches <code>me10-16</code> . This fix from the previous pass is wrong.	Filter on the regenerated question's <code>topic</code> .
MED	topics-varied.js:933	About 14% of order-decimals questions fall back to typed input, and some prompts are already in order.	Reshuffle until unsorted; retry when options are null.
MED	mastery.js:47, difficulty.js	A strong start lowers the difficulty: a new topic is Standard, but after 1–3 answers, even all correct, it drops to Easy.	Treat fewer than 4 attempts as unseen.
MED	Backup.jsx:91	Restore accepts <code>{"name":123}</code> , which wipes the slot. <code>review:{"x":null}</code> crashes the Hub on every load. The active profile is overwritten without asking.	Run <code>parseSlot</code> and per-record validation first, then confirm naming the profile.

SEV	WHERE	PROBLEM	FIX
MED	Game.jsx:247	The answer input has no autoCorrect="off" or autoCapitalize="off" , so iOS autocorrect fixes spelling answers.	Copy the attributes from Quiz.jsx:74.
LOW	session.js:310	The daily challenge misses a subject in 143 of 200 runs.	Include one question per uncovered subject.
LOW	App.jsx:119,156	Results screens understate coins: the +1 per exam answer and the streak reward are not included in the total.	Show the full total.
LOW	vite.config.js, index.html	The app links two manifests (the first isn't precached). Google Fonts is loaded without crossorigin , so it is never cached and there is no font offline. autoUpdate swaps versions silently.	Delete public/manifest.json ; add cacheableResponse:{status:[0,200]} .

Verified correct: tic-tac-toe AI move legality and win detection on 3×3, 4×4 and 5×5; exam sizes and grade bands; passage clusters staying together; topic caps; runner-game respawn and gap sizes.

3 • Answer checking (isCorrect)

SEV	CHILD TYPES	RESULT	STORED / EXPECTED
HIGH	don't, we'll (iOS smart quotes)	Rejected ✗	don't. 290 of 3,340 typed literacy answers contain an apostrophe.
HIGH	9:20, 9.20	Rejected ✗	09:20. 90 time answers are affected, and the prompt's example doesn't show the zero.
MED	-8 (Unicode minus)	Rejected ✗	-8
MED	7 / 8, 3 : 2, 2 : 3	Rejected ✗	7/8, 3:2 (the prompt itself prints "36 : 54")
MED	78p, 126p, £0.50	Rejected ✗	0.78, £1.26, 50p
MED	12 m, 5 kg	Accepted ✓ (false positive)	12 cm, 5 g. Only £/p units are checked.
MED	organize, recognize	Rejected ✗	-ise only, but Oxford -ize is valid British spelling.

SEV	CHILD TYPES	RESULT	STORED / EXPECTED
LOW	28 for "round to 1 d.p.", 1 2, 0x10, 1e2	Accepted ✓ (false positive)	28.0, 12, 16, 100
LOW	90 degrees, 25 percent, 12 cm., 0.5 for 1/2	Rejected ✗	90, 25, 12, 1/2

One pass fixes all of these. Map "→" and "→", collapse spaces around / and :, and zero-pad the hour. Compare times and fractions structurally, convert between p and £, check any unit the child states, accept only `^[+-]?\d*\.\d+$` as a number, and keep trailing zeros for d.p. questions.

4 • Maths content 52,500 generated · 252 static items + 282 generated read by hand

SEV	WHERE	PROBLEM	FIX
CRIT	maths.js:298–302	HARD "scored X out of Y" rounds the true percentage without saying so. "98 out of 150" gives 65% (true value 65.33%). Wrong in 31 of 75.	Choose totals where <code>total·pct/100</code> is a whole number.
CRIT	maths-advanced.js:157	A draft was left in the prompt: "...each term is the previous term squared. 1, 1, 4 ... Actually: the rule is add the previous two terms."	Rewrite as a clean Fibonacci item.
CRIT	maths-advanced.js:282, 347; me1-10	A distractor equals the answer: 5/10 offered for 1/2, and £0.65 for 65p. me1-11 has 10p and £0.10 as two options.	Use misconception distractors (6.5p, £65).
HIGH	topics-varied.js:490	The line graph's "most members" question ties 4–8% of the time; only the first month is keyed as correct.	Re-roll on ties.
HIGH	visual.js:1065; me8-9, me8-13, adv :428/:543/:550	Line graphs have unlabelled points and 3 gridlines at odd values, yet questions ask for exact differences with options 1 apart.	Label points or use gridlines at 5 or 10.
HIGH	topics-varied.js:588	A float is shown as "1000.00000000000001 ml".	Compute <code>amt*want/serves</code> and round.

SEV	WHERE	PROBLEM	FIX
HIGH	topics-varied.js:201,846,963...; challenges.js:102-265	Fixed "she"/"he" with random names gives 1,862 mismatched prompts ("Freya... for his birthday").	Store a pronoun with each name.
HIGH	me1-13, me12-6, me6-9, me10-11, me12-10, me3-17	The visual prints the answer ("Left ? : 35", " ≈ 10 km", the angle labelled 140° , sequences that include the answer).	Remove the answer from each visual.
MED	topics-varied.js:625	"Write it like $\frac{3}{4}$ " accepts only the simplest form; $\frac{2}{4}$ is rejected, and 144 of 450 need simplifying.	Say "simplest form" in the prompt, or accept equivalent fractions.
MED	topics-varied.js:907	The prompt says "three pack sizes" but the table shows four.	Fix the count.
MED	challenges.js:176	The walker sponsors herself (133 of 450).	Exclude the walker from sponsors.
MED	topics-varied.js:864; me7-5	The hint gives the answer away ("Work out 114×10 , then divide by 10"; D's coordinates stated).	Rewrite the hints.
MED	visual.js:910, journeySvg, quadAngleSvg	Visuals contradict the text: "£4.5", the timeline is capped at 6h for 7–9h trips, and the quad shape doesn't match the angles given.	Use toFixed(2) , scale the axis, and draw from the given data.
MED	numericOptions	Distractors are answer ± 1 or $\times 2$, not misconceptions. me2-1's answer is the only option in pence, which gives it away.	Build per-style error distractors.
MED	tiers	best-value EASY needs $\text{£}4.56 \div 8$. me4-4 needs a square root. place-value HARD uses 8 digits (beyond Second level).	Retier these.
MED	scenarios	Unrealistic values: £500–£1200 jackets, £28 pens, cyclists at 60–100 km/h for 8h, paying with £58 (not a note value), 396 pupils in one P7 year group.	Use realistic ranges; pay only with £5/£10/£20/£50 notes.

SEV	WHERE	PROBLEM	FIX
LOW	challenges.js:364, others	Awkward zero and one values: "first 1 hours", "0 h 40 min", HCF of a number with itself, "35 : 35". Only 10 names; 4 duplicate items.	Guard these cases; widen the name pool.
LOW	me3-3, maths-advanced.js:678	Imperial/US framing ("6 feet", Fahrenheit); km/h used for UK road journeys.	Use metric/Celsius, and mph for UK roads.

5 • Literacy content 983 items + 2 passages, all read

SEV	ITEM	PROBLEM	FIX
HIGH	gs540	"Adverbial of time": "before breakfast" is also an option, and also correct.	Replace it with "along the canal".
HIGH	gs530	"Those three dogs": "three" is also a determiner (wc138 teaches this).	Remove "three".
HIGH	ctx28	"The new evidence ___ the theory": "weakened" fits as well as "confirmed".	Add context such as "so she was delighted".
HIGH	gs582	The distractor "I lost my key" also backshifts to "had lost", so it is also correct.	Change the distractor.
HIGH	pv106, pv126	"Someone smashed..." also hides the doer.	Name the doer in that option.
HIGH	gs570	"Aisha lent Freya her book. She had read it" is ambiguous, and co123/co148 teach exactly that.	Rewrite the sentence.
MED	gs591, cl118, cl154, tn113, cl145, gs571, gs638, gs631/gs564, tn124	More than one defensible answer, or teaching that contradicts itself: "light blue" hint points to Adverb, defining/non-defining clauses without context, clause-count convention differs between items, will/must certainty contradicts, "government have" (valid British) marked wrong.	Tighten each item and pick one convention.

SEV	ITEM	PROBLEM	FIX
MED	gs540, gs572, gs616, gs634	They ask about the "underlined" word, but nothing is underlined (no visual, or the visual is suppressed).	Quote the word in the prompt.
MED	-able group, apparent, fair, there, hear, coarse, they're/who's/it's	The hint spells out most of the word ("comfort + -able", "= they are").	Strip the root from the hint.
MED	gs555, 556, 625, 626	Only a bare "im"/"il"/"ir" is accepted; "impatient" or "im-" is marked wrong.	Accept the whole word and hyphen forms.
MED	psgV01-c1, c3	The quoted extract contains only one of the four options.	Don't quote the answer sentence.
MED	sm04, sm54, sm41, government, signature	Factual errors: "stationery contains 'pen'"; hint says 3 syllables in certificate (it has 4); a nonsensical geography clue; "Scottish government meets at Holyrood" (it's the Parliament); "head teacher signature" (missing 's).	Correct the text.
MED	organise/recognise rules	Teaches "UK uses -ise, not -ize", which is false (Oxford spelling).	Accept both spellings.
MED	reg12, reg17, rg106, rg134, reg22	Scots is labelled "slang", against CfE's Scots language policy. rg134's "wasnae ... tae" isn't idiomatic.	Frame it as Scots vs Standard English.
MED	ant04	Running out during a fire drill is called "cowardly", and staying is praised. That is a bad safety message.	Replace the scenario.
MED	passages	Only 2 passages and 6 questions, all literal recall.	Add passages and inference questions.
LOW	tn135, rg131, reg02, ctx14	Weak distractors that are arguably correct.	Replace them.

SEV	ITEM	PROBLEM	FIX
LOW	various	Rule errors: marvel+l+ous, sm51 e-retention reason, gs553 "-in drops", ant24 "innocent" (juries say not guilty), rg113 "might" called obligation. The "-ible when the root is not a word" rule is contradicted by sensible and accessible.	Correct each one.
LOW	co110/co135, tn3/tn118, rg4/rg112 +8	11 exact duplicates, about 30 near-template items, and a pool of about 9 names. rg101 "went a bit mental".	Dedupe, widen names, change the wording.

Method: parallel reviews of UX, function, maths and literacy, then spot-checks against the repo: the Fibonacci draft prompt, percentage rounding, the pronoun templates, `isCorrect` on `9:20/don't/-8/12 m`, and the TimesTable timeout path. All reproduced. The probe scripts are in the session scratchpad. Not covered: a rendered visual/layout pass at phone, tablet and desktop in both themes.